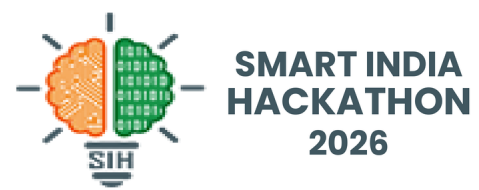
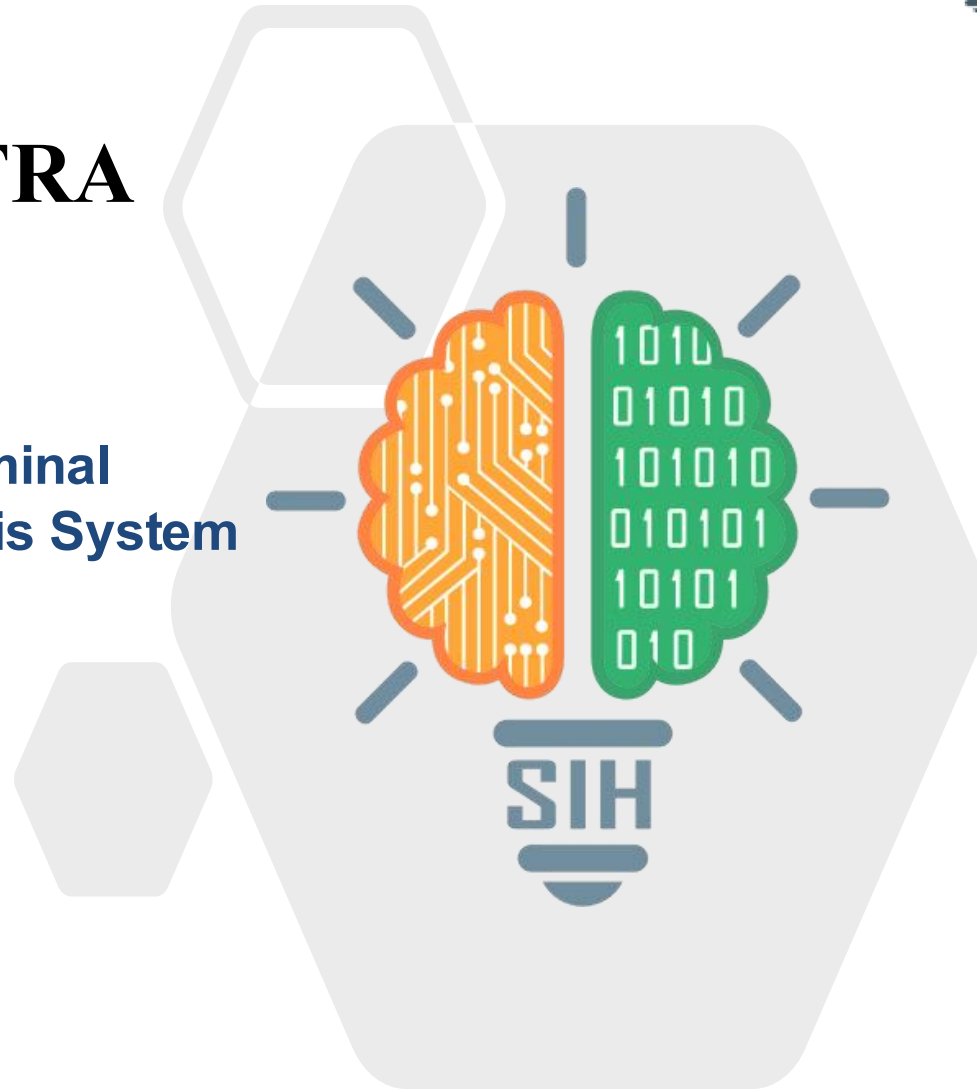


SMART INDIA HACKATHON 2026



NETRA

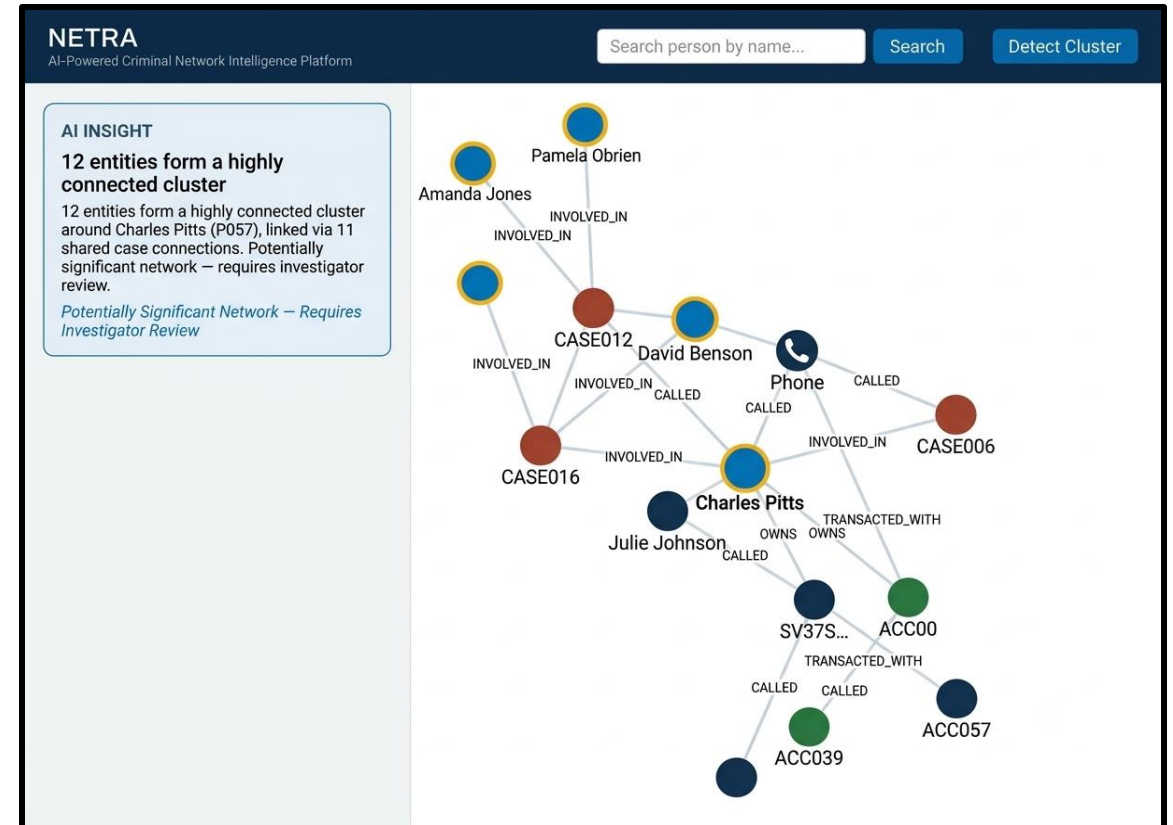
- Problem Statement ID - **SIH26189**
- Problem Statement Title - **AI-Powered Criminal Network Analysis System**
- Theme- **Blockchain & Cybersecurity**
- PS Category- **Software**
- Team ID- **119498**
- Team Name- **NETRA**



“An AI-powered system for discovering hidden relationships and significant networks across heterogeneous investigative data.”

PROPOSED SOLUTION & APPROACH

- **AI Entity Resolution:** Links the same entity across records.
- **Blockchain Evidence Ledger:** AI-generated insight is hashed and recorded on a **permissioned blockchain** with a timestamp and source reference.
- **Network Significance Detection:** Identifies significant **network structures** between **Criminals**.
- **GNN Link Prediction:** **Graph Neural Network** trained on known network patterns predicts likely but undiscovered connections between suspects.
- **Temporal Network Analysis:** Tracks how relationships evolve over time and the **evolution of cases** as well.



Fragmented Data

Entity Intelligence

Relationship Graph

AI / Graph Analysis

Explainable Intelligence



DATA SOURCES



FIR / Case Records



Call Detail Records



Location Data



TECHNOLOGY STACK



Python (FastAPI)



Neo4j (Graphs)



React + Cytoscape.js



Tensor Flow / PyTorch



Docker + Linux

DATA PREPARATION

Data Sources

Ingestion & Validation

AI Entity Extraction

INTELLIGENCE ENGINE

Entity Resolution

Graph + ML Analysis

Temporal Analysis

INVESTIGATION

Dashboard

Explainable Intelligence

Significant Network



data



AI - processing



Graph Data



Analysis



GNN Prediction



Investigator

FEASIBILITY OF NETRA

Technical Feasibility

Built on mature stack: Python, FastAPI, Neo4j, React and Cytoscape.js.

Data Feasibility

Prototype validated using synthetic or anonymized datasets.

Scalability

Graph architecture supports growth from prototypes to large networks.

Operational Feasibility

Human-in-the-loop design keeps AI insights reviewable by investigators.

POTENTIAL CHALLENGES & MITIGATION STRATEGIES

CHALLENGE

MITIGATION

Data Quality

Validation, normalization & schema checks

False Positives

Human-in-the-loop investigator review

Privacy & Security

Encryption, RBAC & audit logs

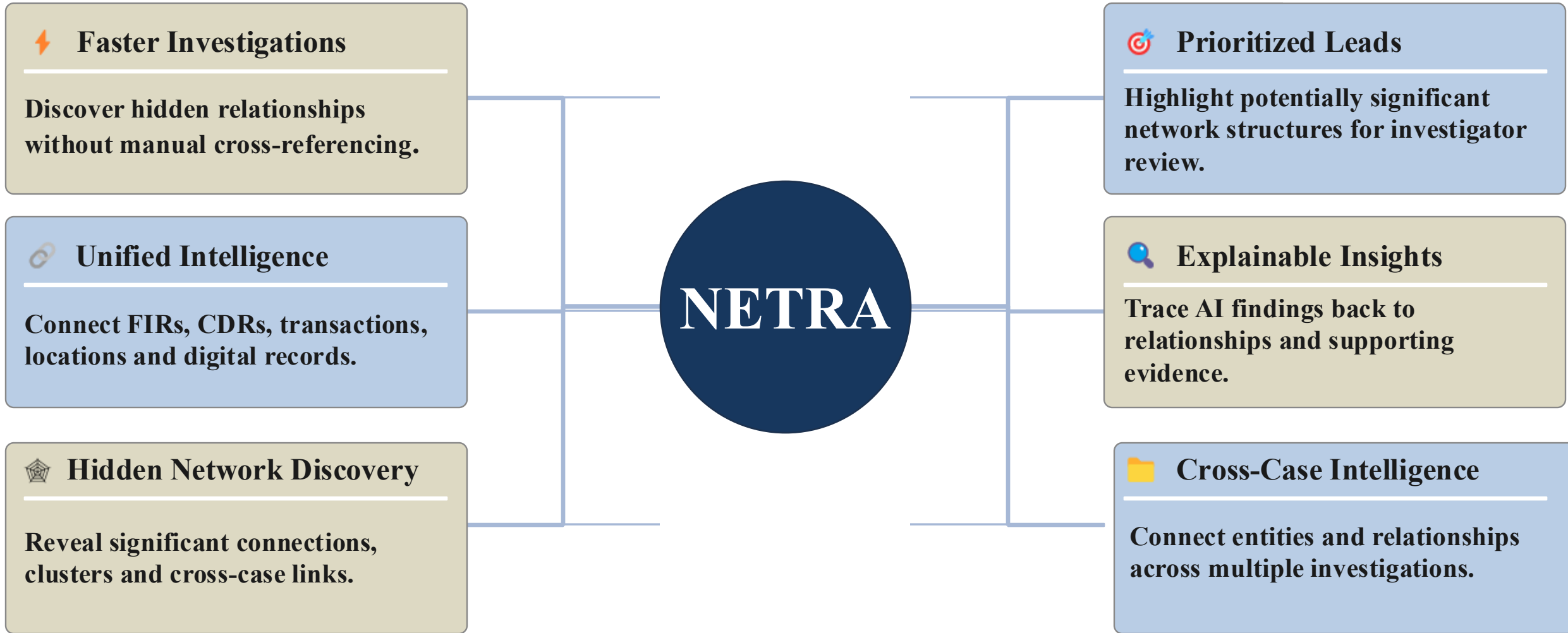
Explainability

Evidence-backed insight tracing

Graph Complexity

Filtering, indexing & optimized queries

VIABILITY — *NETRA can be developed with existing open-source technologies and progressively scaled, subject to data access, authorization and governance.*



“From fragmented records to connected, explainable investigative intelligence.”

RESEARCH FOUNDATION

Graph Analytics

Centrality, communities and network structure

Entity Resolution

Matching entities across inconsistent records

Anomaly Detection

Identifying unusual network patterns

Explainable AI

Evidence-backed and traceable insights

Temporal Analysis

Tracking relationships and network evolution

EXPLORE NETRA



[Source Code](#) • [Documentation](#) • [References](#)

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